

Lab#3

Question 1(a)

```
Editor - C:\Users\National Instruments\Desktop\Untitled2.m
Untitled2.m x +
1 - num=[4 0]
2 - den=[4 3]
3
4 - sys=tf(num,den)
5 - figure(1)
6 - step(sys)
7 - figure(2)
8 - impulse(sys)
9 - figure(3)
10 - bode(sys)

Command Window
New to MATLAB? See resources for Getting Started.

>> Untitled2

num =

    4     0

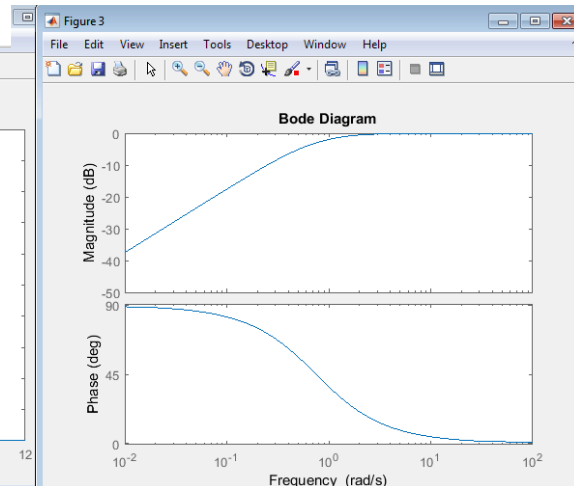
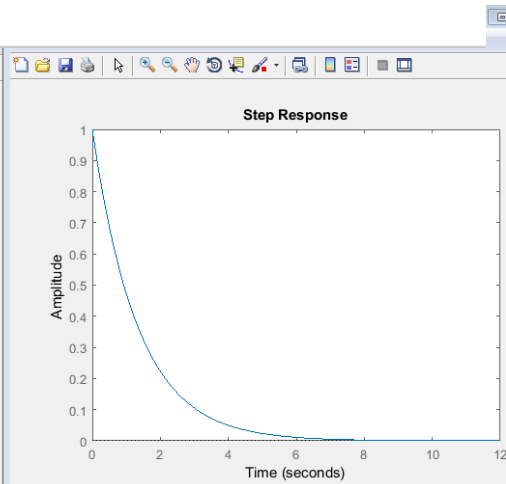
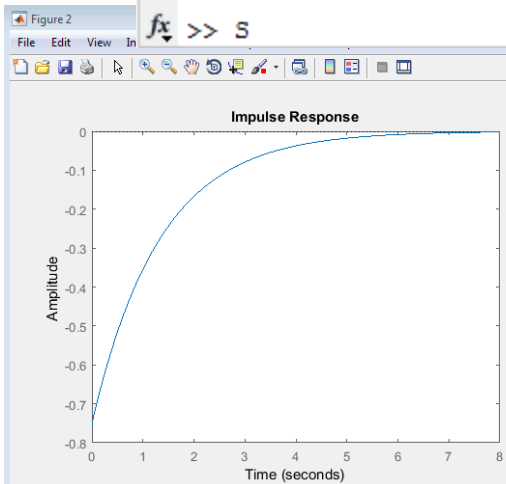
den =

    4     3

sys =

    4 s
    ----
    4 s + 3

Continuous-time transfer function.
```



Question 1(d):

Editor - C:\Users\National Instruments\Desktop\Untitled2.m

```
1 - num=[6 0]
2 - den=[8 6 1]
3 - sys=tf(num,den)
4 - figure(1)
5 - step(sys)
6 - figure(2)
7 - impulse(sys)
8 - figure(3)
9 - bode(sys)
```

Command Window

New to MATLAB? See resources for [Getting Started](#).

```
>> Untitled2

num =

     6     0

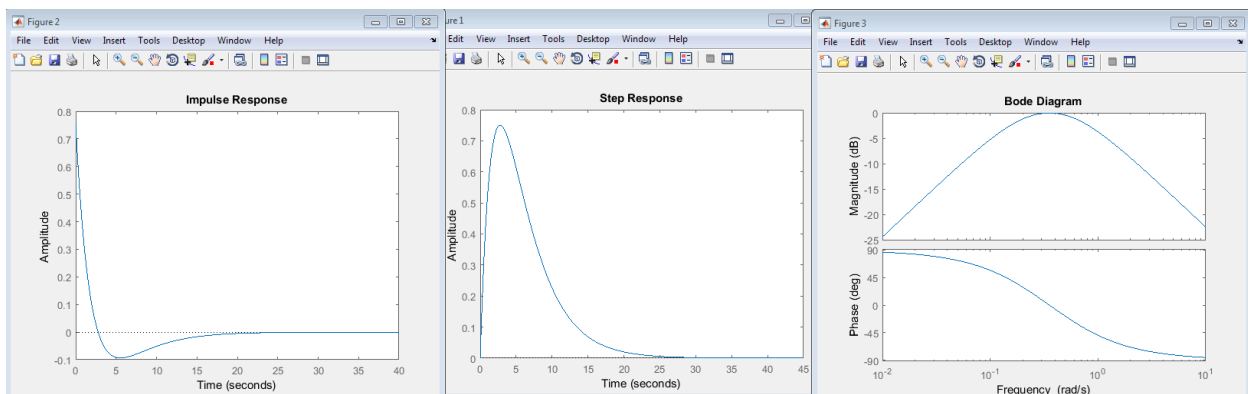
den =

     8     6     1

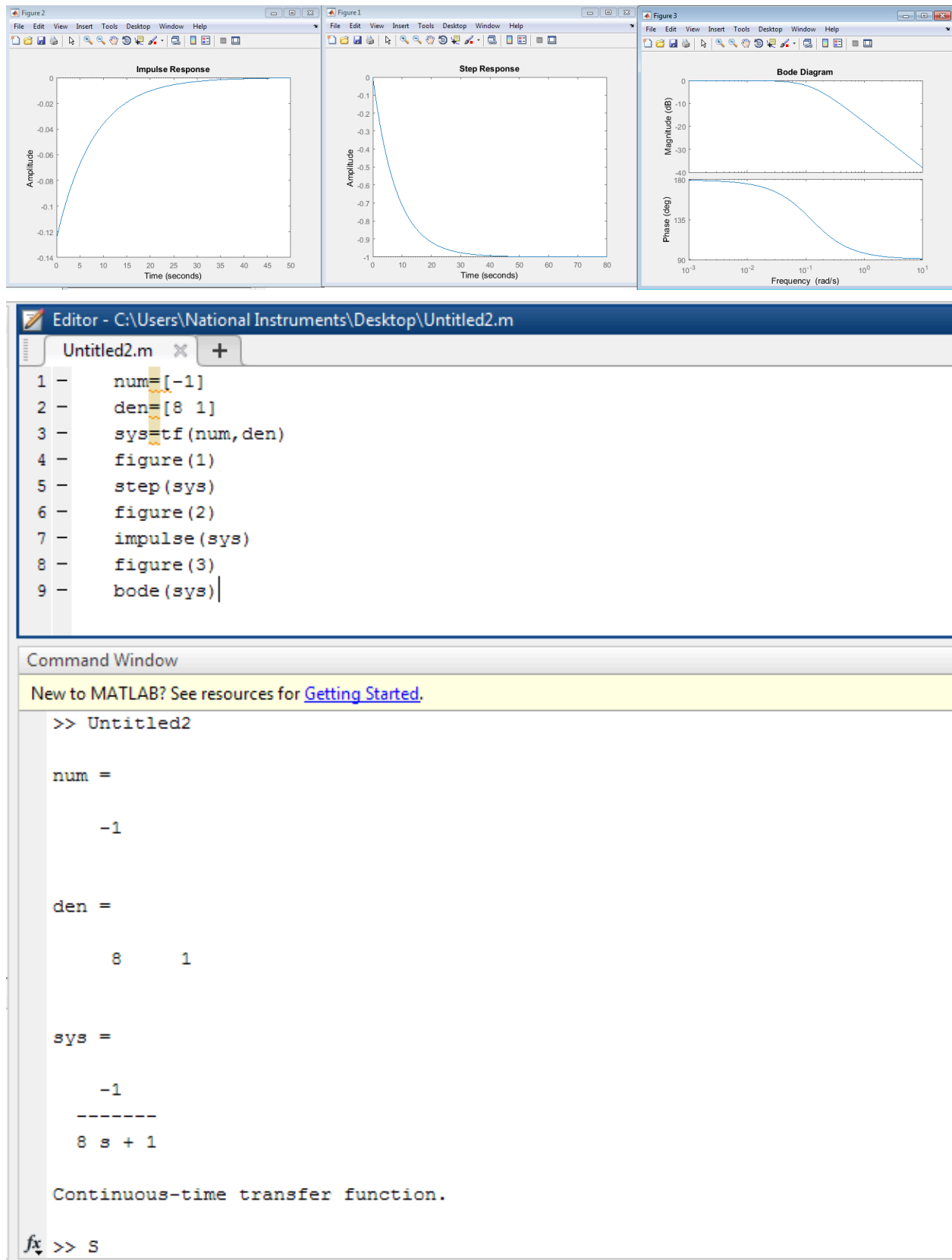
sys =

      6 s
-----
 8 s^2 + 6 s + 1

Continuous-time transfer function.
```



Question 2(c):



Question 2(d):

```
Editor - C:\Users\National Instruments\Desktop\Untitled2.m
Untitled2.m x +
1 - num=[2]
2 - den=[1]
3 - sys=tf(num,den)
4 - figure(1)
5 - step(sys)
6 - figure(2)
7 - impulse(sys)
8 - figure(3)
9 - bode(sys)

Command Window
New to MATLAB? See resources for Getting Started.

sys=tf(num,den)

>> Untitled2

num =

    2

den =

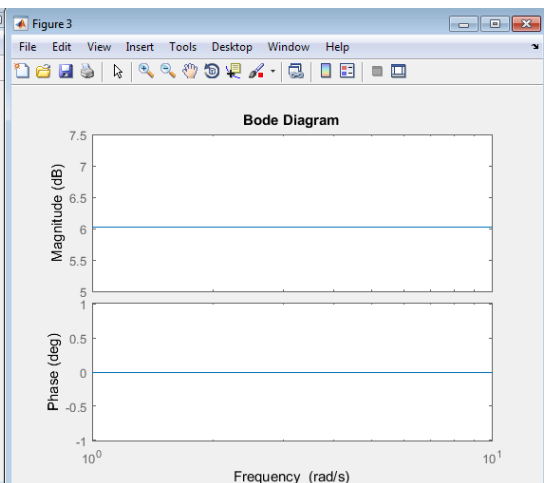
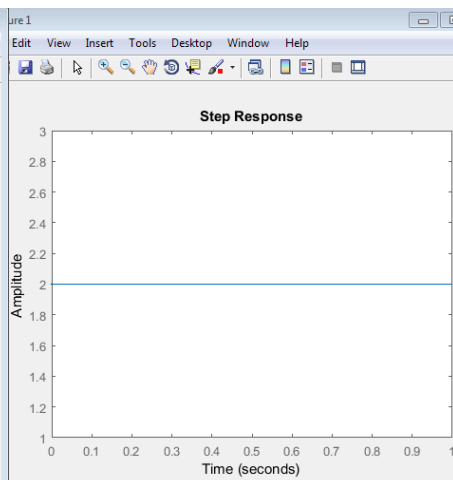
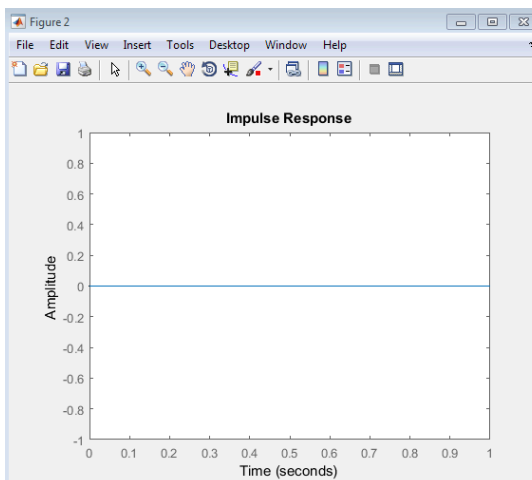
    1

sys =

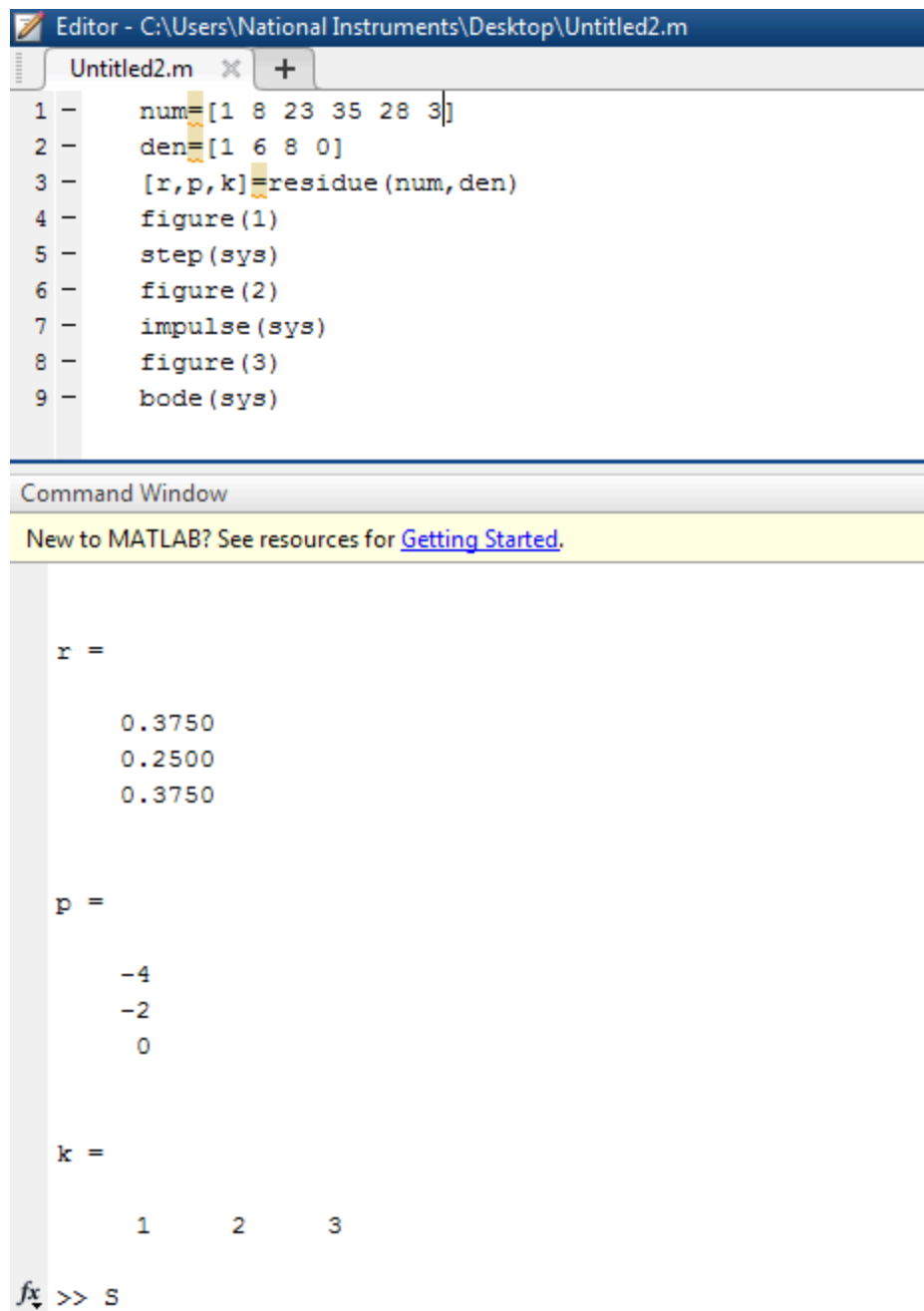
    2

Static gain.

fx >> s
```



Exercise 1:



The image shows a MATLAB environment with an Editor window and a Command Window. The Editor window, titled 'Editor - C:\Users\National Instruments\Desktop\Untitled2.m', contains a script with 9 lines of code. The Command Window, titled 'Command Window', displays the output of the 'residue' function, showing the residues (r), poles (p), and direct terms (k) for the given numerator and denominator polynomials. A yellow banner at the top of the Command Window reads 'New to MATLAB? See resources for [Getting Started.](#)'

```
Editor - C:\Users\National Instruments\Desktop\Untitled2.m
Untitled2.m  x  +
1 - num=[1 8 23 35 28 3]
2 - den=[1 6 8 0]
3 - [r,p,k]=residue(num,den)
4 - figure(1)
5 - step(sys)
6 - figure(2)
7 - impulse(sys)
8 - figure(3)
9 - bode(sys)

Command Window
New to MATLAB? See resources for Getting Started.

r =

    0.3750
    0.2500
    0.3750

p =

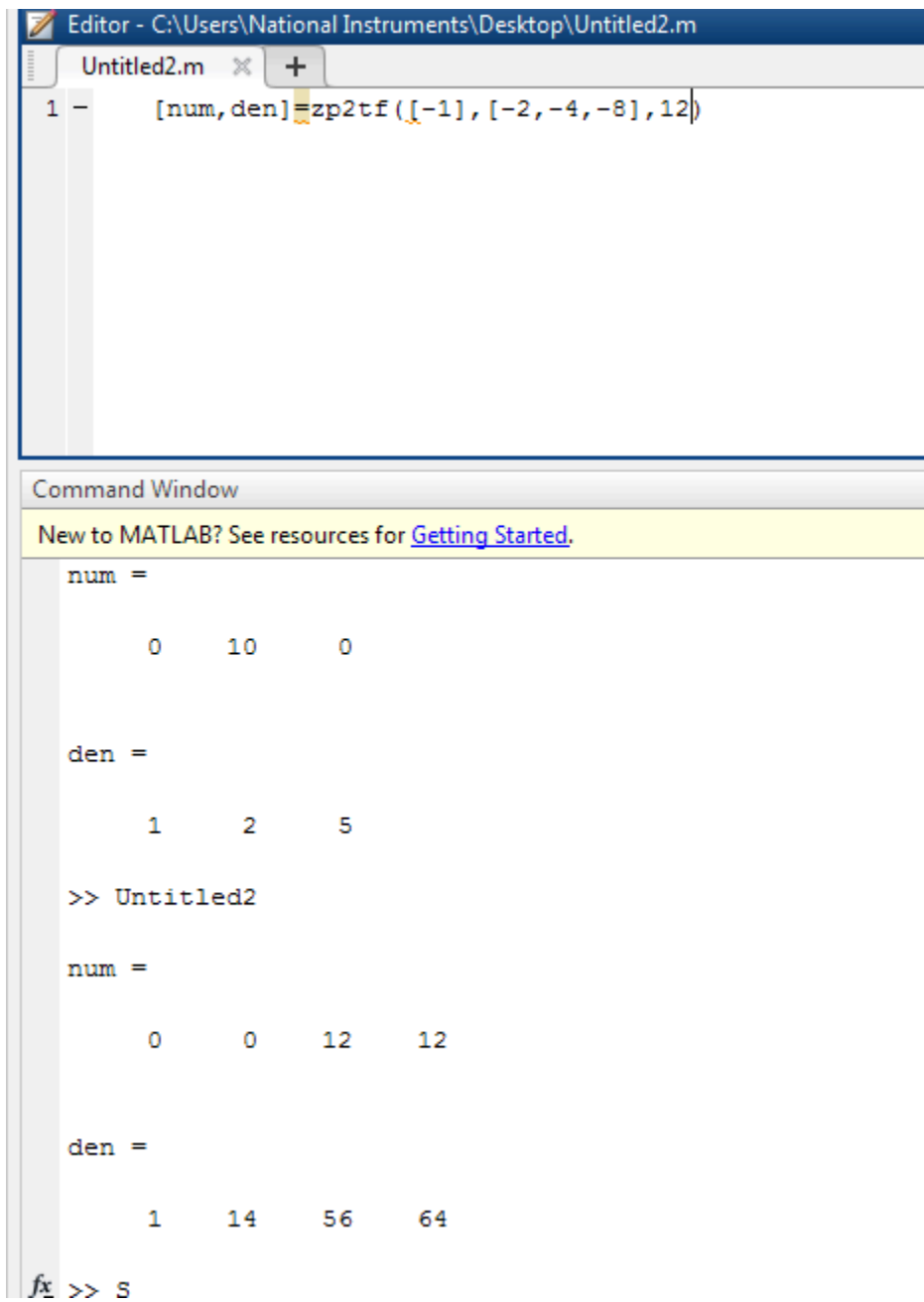
   -4
   -2
    0

k =

     1     2     3

fx >> s
```

Exercise 2:



The image shows a MATLAB environment. The Editor window at the top displays a script named 'Untitled2.m' with the following code: `1 - [num,den]=zp2tf([-1],[-2,-4,-8],12)`. The Command Window at the bottom shows the execution of this script. It first displays the zeros and poles of the system, then the transfer function in polynomial form, and finally the transfer function in factored form.

```
Editor - C:\Users\National Instruments\Desktop\Untitled2.m
Untitled2.m x +
1 - [num,den]=zp2tf([-1],[-2,-4,-8],12)

Command Window
New to MATLAB? See resources for Getting Started.

num =
    0    10    0

den =
    1    2    5

>> Untitled2

num =
    0    0   12   12

den =
    1   14   56   64

fx >> s
```

Exercise 3:



The image shows a MATLAB environment with an Editor window and a Command Window. The Editor window, titled 'Editor - C:\Users\National Instruments\Desktop\Untitled2.m', contains a single line of code: `[num,den]=zp2tf([-1;-2],[0,-4,-6],[5])`. The Command Window, titled 'Command Window', displays the output of running the script 'Untitled2'. It shows the numerator coefficients 'num = 0 5 15 10' and the denominator coefficients 'den = 1 10 24 0'.

```
Editor - C:\Users\National Instruments\Desktop\Untitled2.m
Untitled2.m
1 - [num,den]=zp2tf([-1;-2],[0,-4,-6],[5])

Command Window
New to MATLAB? See resources for Getting Started.

>> Untitled2

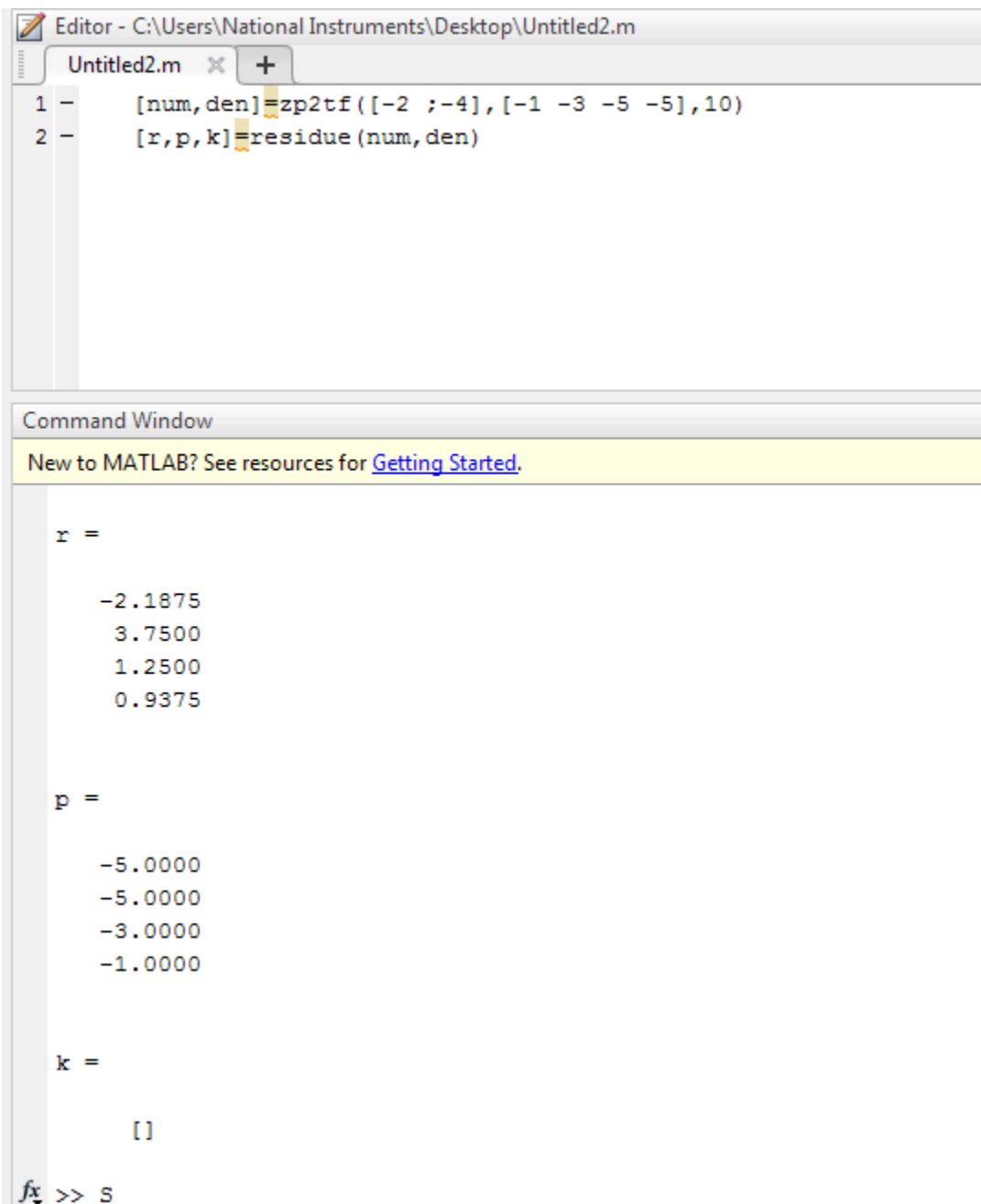
num =

    0     5    15    10

den =

     1    10    24     0
```

Exercise 4:



The image shows a MATLAB environment with an Editor window and a Command Window. The Editor window, titled 'Editor - C:\Users\National Instruments\Desktop\Untitled2.m', contains two lines of MATLAB code. The Command Window, titled 'Command Window', displays the output of the code, showing the residue decomposition of a transfer function. It includes a yellow banner for new users and the variable assignments for r, p, and k.

```
Editor - C:\Users\National Instruments\Desktop\Untitled2.m
Untitled2.m
1 - [num,den]=zp2tf([-2;-4],[-1 -3 -5 -5],10)
2 - [r,p,k]=residue(num,den)

Command Window
New to MATLAB? See resources for Getting Started.

r =

    -2.1875
     3.7500
     1.2500
     0.9375

p =

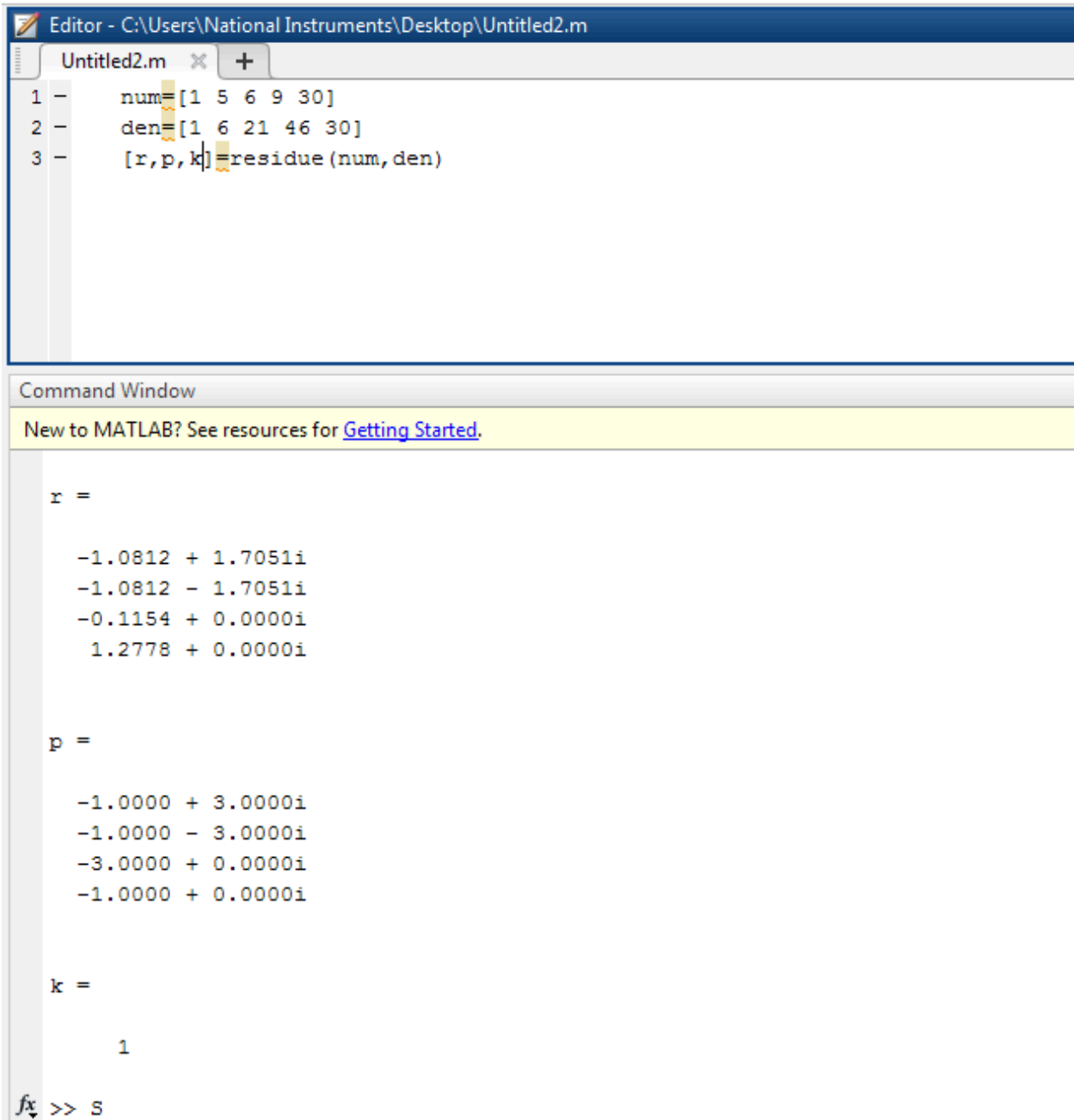
    -5.0000
    -5.0000
    -3.0000
    -1.0000

k =

     []

fx >> s
```


Exercise 5:



The image shows a MATLAB environment with an Editor window and a Command Window. The Editor window, titled 'Editor - C:\Users\National Instruments\Desktop\Untitled2.m', contains three lines of MATLAB code. The Command Window, titled 'Command Window', displays the output of the code, including a message for new users and the numerical results for variables r, p, and k.

```
Editor - C:\Users\National Instruments\Desktop\Untitled2.m
Untitled2.m
1 - num=[1 5 6 9 30]
2 - den=[1 6 21 46 30]
3 - [r,p,k]=residue(num,den)

Command Window
New to MATLAB? See resources for Getting Started.

r =

-1.0812 + 1.7051i
-1.0812 - 1.7051i
-0.1154 + 0.0000i
1.2778 + 0.0000i

p =

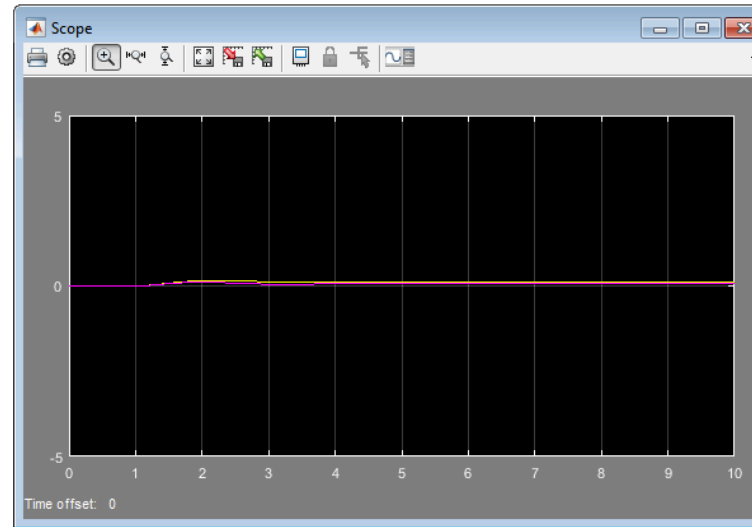
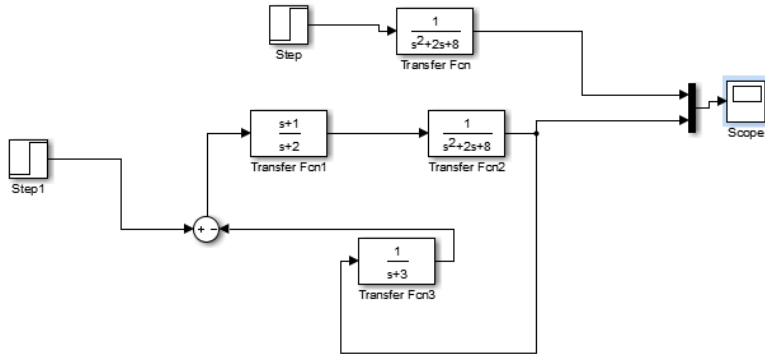
-1.0000 + 3.0000i
-1.0000 - 3.0000i
-3.0000 + 0.0000i
-1.0000 + 0.0000i

k =

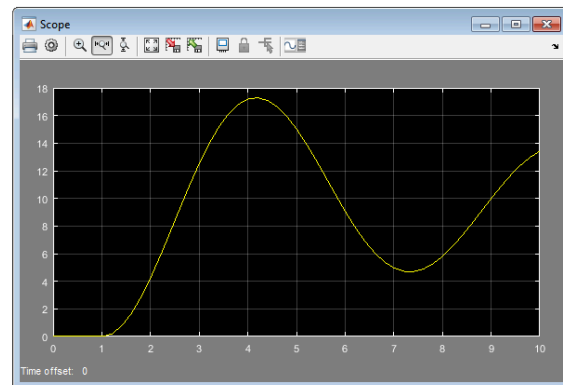
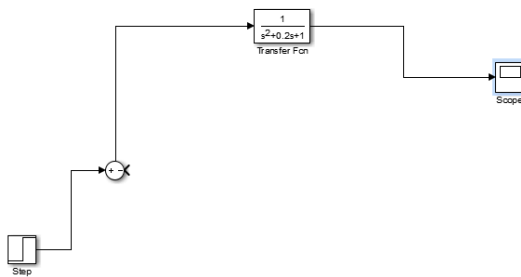
1

fx >> s
```

3.3 - Exercise 1:



3.3 - Exercise 2 (Part 1):



3.3 - Exercise 2 (Part 2):

